AI Music Generation Using LSTM

An AI-powered music generation project that learns patterns from MIDI files and generates new symbolic music using a Long Short-Term Memory (LSTM) neural network.

The notebook uses the MAESTRO MIDI dataset, extracts musical note and chord events, trains a TensorFlow/Keras model, and exports generated music as MIDI and (if conversion succeeds) WAV audio.

Project Overview

This project demonstrates an end-to-end symbolic music generation workflow:

1. Set up the Python environment and required libraries.
2. Download and extract the MAESTRO MIDI dataset.
3. Parse MIDI files and extract note/chord events.
4. Explore the most frequent musical events with a chart.
5. Convert musical events into numeric sequences.
6. Train an LSTM model to predict the next musical event.
7. Generate a new sequence using the trained model.
8. Export the sequence as a MIDI file and render it to WAV audio.

Features

- Downloads and processes MIDI files from the MAESTRO dataset.
- Selects piano parts when available; otherwise, uses the first available part.
- Represents notes and chords as symbolic tokens.
- Visualizes the 15 most frequent musical events.
- Trains an LSTM-based sequence prediction model.
- Uses model checkpointing and early stopping.
- Generates a sequence of new musical events.
- Exports generated music to MIDI.
- Converts MIDI to WAV with TiMidity when available.
- Plays the generated audio in Google Colab.

Technologies Used

- Python
- TensorFlow / Keras
- NumPy
- Pandas
- Matplotlib
- music21

- tqdm
- pretty_midi
- TiMidity

Requirements

The notebook is designed for **Google Colab** because it uses `/content` paths and `google.colab.files`. An internet connection is required to download the dataset and install dependencies.

The notebook installs the Python packages with:

```
pip install music21 pretty_midi tensorflow numpy pandas matplotlib tqdm
```

It also installs TiMidity for MIDI-to-WAV rendering:

```
apt-get update -qq
apt-get install -y -qq timidity
```

These commands are included in the notebook.

How to Run

1. Open the notebook in Google Colab.
2. Run the environment setup cell.
3. Run the dataset download and extraction cell.
4. Run the MIDI processing and event extraction cells.
5. Review the musical-event frequency chart.
6. Run the sequence preparation and model-building cells.
7. Train the model.
8. Review the training loss and accuracy plots.
9. Generate a new musical sequence.
10. Create the MIDI file and render it to WAV.
11. Play the generated audio in the notebook or download the output files.

Model Architecture

The TensorFlow/Keras model contains:

- An input layer for musical-event sequences.
- Two LSTM layers with 256 units each.
- Dropout layers to help reduce overfitting.
- A dense hidden layer with 128 units and ReLU activation.
- A softmax output layer to predict the next event from the learned vocabulary.

The model uses the Adam optimizer and categorical cross-entropy loss. Training is configured for up to 30 epochs with a batch size of 64, with checkpointing and early stopping enabled.

Configuration

Setting	Default value	Purpose
MAX_FILES	Up to 50	Maximum number of MIDI files selected
SEQUENCE_LENGTH	100	Previous events used as model input
EPOCHS	30	Maximum training epochs
BATCH_SIZE	64	Training examples per batch
GENERATE_LENGTH	300	Number of events generated

These values can be adjusted according to the available runtime and experiment goals.

Output Files

The notebook creates these files in the Colab `/content` directory:

- AI_Generated_Music.mid — generated symbolic music in MIDI format.
- AI_Generated_Music.wav — rendered audio, if TiMidity conversion succeeds.
- best_music_model.keras — saved model checkpoint.

The notebook prints commands for downloading these files from Google Colab.

Visualizations

The notebook produces plots for:

- The 15 most frequent musical events in the selected dataset.
- Training loss across epochs.
- Training accuracy across epochs.

These visualizations help inspect the dataset and monitor the training process.

Limitations

- Only up to 50 MIDI files are processed by default, so output may be repetitive or limited.
- The model predicts symbolic events; it does not directly generate recorded audio.
- Choosing the highest-probability event each time can produce repetitive sequences.
- Training accuracy alone does not measure the musical quality of a composition.
- MIDI-to-WAV conversion depends on TiMidity and its sound configuration.
- The notebook uses Colab-specific paths and download utilities, so local execution may require changes.
- Dataset processing and model training can take time and memory.